MILITARY SPECIFICATION

CARBON MICROPHONE M-15( )/UR

1. SCOPE

1.1 This specification covers one type of noise-cancelling, hand-held, carbon microphone, designated Carbon Microphone M-15( )/UR.

1.1.1 In the type designation above, the parenthesis will be either removed or replaced by a letter. This information will be furnished after each contract has been awarded upon application by the contractor to the Procuring Agency. The complete nomenclature or type number will be used on nameplates, shipping records, and in instruction books, as applicable.

2. APPLICABLE SPECIFICATIONS, DRAWINGS, AND OTHER PUBLICATIONS

2.1 The following specifications, drawings, and other publications, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein.

2.1.1 Specifications:

Federal:

QQ-M-151	Metals; General Specification For Inspection of
QQ-P-416	Plating; Cadmium (Electrodeposited)

Military:

JAN-C-76	Cable (Hook-Up Wire), Electric, Insulated Radio and Instrument
JAN-M-745	Moisture-Resistance-Test, For Material Items Used in Electronic, Communication, and Electrical Equipment
JAN-P-642	Plugs, Telephone
MIL-F-5030	Felt; Hair
MIL-E-5135	Electronic-Standard-Items; Part, Material and Process
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)
MIL-F-5656	Felt; Wool (For Aircraft Applications)
MIL-M-5658	Mildew Proofing of Fabrics, Threads and Cordages; Copper Processes For
MIL-P-6859	Plating, Nickel
MIL-P-6871	Plating, Chromium

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Air Force-Navy Aeronautical:

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (For Aluminum Alloys)
AN-P-52	Plating; Zinc
AN-S-126	Screw-Threads; Standard Aircraft

U. S. Army:

71-4945	Rubber Substitutes, For Use In Cords, Cordages, and Cables
71-886	Cordage CO-122-A and CO-122-B

U. S. Air Force:

7200	Electronic Equipment; Aircraft, Preservation, Packaging, Packing and Marking For Shipment Of

2.1.2 Drawings:

Air Force-Navy Aeronautical:

AN659	Terminal-Electrical, Crimp Style, Copper

U. S. Air Force:

SC-D-391	Switch SW-109 Assembly & Details
SC-D-2268	Microphone T-17 General Assembly
SC-B-2271	Microphone T-17, Circuit Label
SC-D-5975	Carbon Microphone, Test Circuits
SC-C-10574	Microphone T-17 Cord Assembly & Details
SC-D-10599	Microphone Test Equipment ARL-174-A, Driver Circuit and Calibration Information
49B12538	Spectrum-Simulated Aircraft Noise

2.1.3 Other Publications:

Office of Scientific Research and Development Report No. 3802

(Copies of this specification, Federal, Military, Air Force-Navy Aeronautical, U. S. Air Force, and U. S. Army specifications and drawings, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies. Copies of Office of Scientific Research and Development Report No. 3802 can be obtained from the Department of Commerce, Washington, D. C.)

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3. REQUIREMENTS

3.1 Assembly.- The microphone shall be equivalent to Microphone T-17-( ), as shown on Drawing No. SC-D-2268, with the major change being use of noise-cancelling Microphone Element M-14( )/UR instead of a straight pressure type microphone element. The microphone element shall include the mouthpiece as a part thereof. The assembly shall also include a microphone cover.

3.2 Materials.- The materials, component parts, and mechanical assemblies used in the construction of this microphone, but not specified in detail, shall be of a quality and nature entirely consistent with the proposed use and specified performance of the microphone.

3.2.1 Use of Electronic Standard Parts, Materials, and Processes.- The use of electronic standard parts, materials, and processes shall be in accordance with Specification MIL-E-5135 with the following exception:

3.2.1.1 Substitution of Parts, Materials and Processes.- If a manufacturer, either as a bidder or a Contractor, desires to substitute another part, material, or process where the specification and drawings call for a specific part, material, or process; or where the particular part, material, or process has previously been approved for use in the microphone, he shall first submit to the Procuring Agency concerned a statement to that effect, describing the proposed substitution and the reason therefor, and shall submit evidence that such a substitution is at least the equal of the specific part, material, or process specified or previously approved. At the discretion of the Procuring Agency samples may be required which will demonstrate by test the suitability of the proposed substitution.

3.2.2 Screws.- Screw threads shall be in accordance with Specification AN-S-126. Sizes smaller than No. 3 shall not be used, except on small assemblies not intended to be taken apart, or in cases when the design requirements cannot otherwise be met. The lengths of screws shall be such as to provide a minimum of 1-1/2 threads and a maximum of 1-1/2 threads plus 1/8 inch beyond the nut or joining components. Thread engagements of screws shall provide a minimum of 1/8 inch effective thread in tapped parts where an assembly is frequently dissembled. A minimum of three complete threads shall be used in aluminum, aluminum alloys, and other soft materials or non-metals. A minimum of two complete threads shall be used in stronger materials.

3.2.3 Locking Devices and Lock Washers.- All screw-thread assemblies shall be made vibration-proof by the use of lockwashers, locknuts, or by means of mechanical staking. Lockwashers shall be plated bronze, corrosion-resisting steel, cadmium-plated steel, or zinc-plated steel.

3.2.4 Soldering.- Soldering shall be such that both a positive electrical and a strong mechanical connection is assured. Unless specifically approved by the Procuring Agency, no assembly shall depend solely on soft solder for mechanical strength.

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3.2.5 Terminals.- Terminals attached to parts shall be mechanically strong and shall not bend or break when soldering or unsoldering connections thereto. Terminals attached to wires may be of either the solder or solderless type. The solderless terminals shall be in accordance with Drawing AN659.

3.2.6 Wires.- Hookup wire used in the microphone shall be in accordance with the requirements of Specification JAN-C-76. Wiring shall be distinctly color coded in order to facilitate testing. Insulation on conductors shall be integral with the conductor, except that insulation of the sleeving type may be used provided its length is not greater than 4 inches and is sufficient to prevent slippage along the conductor to such an extent as to expose a section of the conductor that should be insulated. All conductor insulation shall be of the non-combustible or slow-burning type.

3.2.7 Carbon and Carbon Cavity:

3.2.7.1 The carbon granules shall be of a grade, quality, size, and shape suitable in every respect for this particular application.

3.2.7.2 All contact surfaces within the carbon cavity shall be suitably plated with gold.

3.2.7.3 The carbon granules shall be retained positively in the microphone element in such a manner that it is impossible for them to leak from the carbon cavity during normal service use, or under any of the conditions specified herein:

3.2.8 Wiring.- All wiring connecting to terminals shall be securely fastened to the terminals by crimping the terminals firmly upon the wire or the wire upon the terminals before soldering, except in cases where crimping is impractical. In no case shall electrical connections depend upon wires, lugs, terminals, and the like clamped between a metallic member and an insulating material of other than a ceramic or vitric nature.

3.2.9 Cotton and Felt.- Felt shall be in accordance with Specification MIL-F-5656 or MIL-F-5030, and in addition shall be treated to be water repellent and to be fungus resistant, so as to meet tests and requirements of Specification MIL-M-5658, except that the treating compounds and processes need not be restricted to those described in that specification. Cotton shall be treated in accordance with Specification MIL-M-5658.

3.2.10 Switch SW-109:

3.2.10.1 Spring Material.- The sheet metal springs used in this switch shall be formed with the grain parallel to the length of the spring, and shall have a spring temper suitable for this application.

3.2.10.2 Action.- The switch shall normally be in the open position. Using a microphone having the back section of the housing removed, with the switch held in the extreme "ON" position, a force of not less than 12 ounces,

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as measured on a standard Graham gauge, shall be required to pull the lowest spring contact (Item 20, Drawing No. SC-D-391, Assembly View) sufficiently to open the electrical circuit between Items 19 and 20 (Drawing No. SC-D-391). This shall be measured by applying a pulling force perpendicular to the spring leaf and approximately on the center line of the contact.

3.2.10.3 Contact Spacing and Sequence.- The width of the gaps between each set of contacts shall be not less than .035 inch when the switch is in the normal (open) position. The microphone contacts (those on Items 17 and 18, Drawing SC-D-391) shall close before the control circuit contact (Item 21, Drawing SC-D-391). The over-travel of the switch, after the control circuit contacts close, shall be not less than .020 inch. Furthermore, once the microphone and control circuits are closed, they shall remain closed throughout the over-travel of the switch.

3.2.10.4 Life Under Repeated Stress.- The switch shall be capable of withstanding at least 100,000 switching cycles. Each switching cycle shall consist of one complete depression of the switch (including over-travel) and return to normal (open) position at a rate of approximately 60 cycles per minute.

3.2.11 Cordage.- Cordage designated as Cordage CO-122-A or B on the applicable drawings shall be in accordance with Specification No. 71-886, with the following additions or exceptions:

3.2.11.1 Central Cores, Fillers, Binders, Etc.- All cotton or other textiles used as a central core, filler, binder, etc., shall be mildew-proof treated in accordance with Specification MIL-M-5658. The impregnating material shall not affect the electrical or physical characteristics of the cordage.

3.2.11.2 Temperature Requirements:

3.2.11.2.1 Operating Range.- The cordage shall be flexible and resilient throughout the temperature range of -55°C to 85°C.

3.2.11.2.2 Cold Bend.- The cordage shall be capable of meeting the cold bend requirements of Specification No. 71-4945 with the cold chamber temperature maintained at -55°C, ±2°C.

3.2.11.2.3 Flexing.- The cordage shall be capable of withstanding at least 7000 flexing cycles without showing any evidence of:

(1) Damage to the jacket of the cordage.
(2) Damage to the insulation of the individual conductors.
(3) Electrical discontinuity.

3.2.12 Plug.- Plug PJ-068 per Specification JAN-P-642 shall be used in lieu of Plug PL-68 specified on Drawing Nos. SC-D-10574 and SC-D-2268.

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3.3 Design.- The microphone shall be a noise canceling, carbon microphone. In general, the design and assembly of the microphone shall be such as to have inherently stable mechanical, electrical, and acoustical characteristics. It shall not be necessary to condition the microphone by any special procedure before it is ready for use.

3.4 Connections.- The microphone shall be connected as shown on Drawing Nos. SC-B-2271 and SC-D-2268 except that a 6800 mmfd, 200 volt capacitor, Type 191P68202 as made by the Sprague Electric Co. or an 8200 mmfd, 200 volt capacitor, Type XF1870, as made by the Gudemin Co., or equivalents, shall be connected across the microphone unit terminals.

3.5 Response Characteristics.- The response of the microphone measured as specified herein shall be expressed in decibels relative to a zero reference level of 1.0 millivolt. The response shall be such as to meet the following requirements when measured with the grid of the microphone unit in a vertical plane.

3.5.1 Response Limits.- The response shall fall within the limits of 45.0 and 53.0 decibels over the 300 to 4000 cycles per second range, inclusive.

3.5.2 Average Response.- The arithmetical average of the test frequency response values over the 300 to 4000 cycles per second frequency range shall be between the limits of 47.0 and 51.0 decibels, inclusive.

3.5.3 Peaks and Dips.- The response characteristic shall be substantially free from sharp peaks and dips over the 300 to 4000 cycles per second frequency range.

3.6 Distortion.- Distortion shall be held to the lowest practicable minimum.

3.7 Microphone Resistance.- The resistance of the microphone shall be between 60 and 120 ohms.

3.8 Burning.- The output of the microphone due to burning shall be kept to the lowest practicable minimum, and in no case shall it exceed 1.0 millivolt.

3.9 Aging Characteristics.- The design and construction of and the materials used in the microphone shall be such that the microphone shall maintain its original performance characteristics in use or in storage. The microphone shall comply with the following requirements when artificially aged as specified in 4.5.2.3.

3.9.1 Change in Resistance Due to Aging.- The resistance of the microphone shall not increase more than 75 percent of its original resistance as a result of the prescribed aging procedure.

3.9.2 Change in Response Due to Aging.- The average response of the microphone over the 300 to 4000 cycles per second frequency range shall not change more than 3.0 decibels as a result of artificial aging.

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3.9.3 Quality of Transmission After Aging.- The effectiveness of the microphone in transmitting speech intelligibly under high ambient noise levels shall not be appreciably affected as a result of artificial aging.

3.10 Anti-Packing:

3.10.1 Surges.- The design of the microphone shall be such that by rapidly switching it in and out of the test circuit shown in Drawing No. SC-D-5975, Figure 5, for 50 times, at a rate of about 25 times per minute, will not result in an average sensitivity loss of more than 4.0 decibels.

3.10.2 Overload Current.- The microphone shall be capable of retaining its response characteristics within 3 decibels at 1000 cycles per second immediately after being subjected to a direct current excitation current of 80 milliamperes for one hour.

3.10.3 Extended Operation.- The microphone shall be capable of transmitting intelligible speech immediately after being subjected to an excitation current of 50 milliamperes for 8 hours.

3.11 Positional Operation.- The microphone shall provide satisfactory communication when used in any position (vertical, horizontal, and the like). The microphone shall show no pronounced change in loudness or speech quality when operated in the various positions.

3.12 Articulation.- The average word articulation score of the microphone shall be either 80 percent or more or at least 20 percent higher than that of a standard Microphone T-17 (maintained at Air Development Force, Wright-Patterson Air Force Base, Dayton, Ohio) when tested under noise-to-quiet conditions using a noise level of 120 db at any altitude up to 15,000 feet (approximately 16.7 inches of mercury).

3.13 Interchangeability.- Mechanical and electrical interchangeability shall exist between like assemblies, sub-assemblies, and replaceable parts, regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not necessarily mean identity, but requires that a substitution of such like assemblies, sub-assemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the microphone.

3.14 Dimensions.- Dimensions and tolerances not specified in detail shall be as close as is consistent with best shop practices. Where dimensions and tolerances affect the interchangeability, operation or performance of the microphone, they shall be held or limited accordingly.

3.15 Weight.- The weight of the microphone shall not exceed one pound.

3.16 Service Conditions.- The microphone shall show no mechanical damage and shall provide satisfactory communication performance when operated under any possible combination of the service conditions specified herein. It shall be so

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designed and constructed that subjection to any or all of the specified service conditions will result in no serious change in response curve shape, output level, distortion or intelligibility of transmitted speech. Where any doubt exists as to the seriousness of any change resulting from exposure to service simulating conditions, word articulation tests shall be relied upon to determine the seriousness of the change.

3.16.1 Service Conditions (Atmospheric).- The microphone shall be designed and constructed so that no fixed part or assembly shall become loose, no moving part or assembly become undesirably free or sluggish in operation, and no degradation be caused in the performance below that specified herein because of operation or storage in ambient conditions as follows:

3.16.1.1 Equipment Operating.- Operation of the microphone under any of the following conditions or natural combination thereof:

3.16.1.1.1 Temperature:

(1) Continuous operation of the microphone over the range of -55° to 55°C, (-67° to 131°F) shall be satisfactory. The ambient temperature may remain approximately constant for long periods of time or may vary at a rate as high as 1°C (1.8°F) per second.

(2) Operation of the microphone at 71°C (160°F) for a continuous period of 30 minutes after non-operating stabilization at 71°C (160°F) shall be satisfactory.

3.16.1.1.2 Humidity.- Operation shall be at relative humidity ranging up to 100 percent at temperatures up to 50°C (122°F) for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the microphone in the form of both water and frost.

3.16.1.1.3 Altitude.- Operation shall be at barometric pressures ranging from 30 inches of mercury down to 16.7 inches of mercury (approximately an altitude of 15,000 feet). The pressure may remain constant or may vary at a rapid rate.

3.16.1.2 Equipment Non-Operating.- The following conditions and any natural combination thereof are those to which the microphone may be subjected, while inoperative. The microphone shall sustain no degradation in performance below that specified herein after having been returned to the operating range service conditions for a period of one hour prior to any operation or testing.

3.16.1.2.1 Temperature.- Ranging from -65°C to 85°C (-85°F to 185°F); the temperature may remain constant or vary at a rate as high as 1°C (1.8°F) per second.

3.16.1.2.2 Altitude.- Barometric pressures shall range from 30 inches of mercury down to 3.4 inches of mercury (approximately an altitude of 50,000 feet)

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and may remain constant for long periods of time or may vary at a rapid rate. Suitable provision shall be made in the design to equalize the pressure between the inclosed cavities of the microphone and the outside atmosphere when the microphone is subjected to changes in altitude.

3.16.1.2.3 Humidity.- The humidity range shall be up to 100 percent relative, at temperatures up to 50°C (122°F) for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the microphone in the form of water and frost.

3.16.2 Service Condition (Mechanical).- The microphone shall be designed and constructed so that no fixed part shall become loose and no movable part shall become shifted in setting, position or adjustment because of operation or storage under the following conditions:

3.16.2.1 Vibration.- The response requirements of the microphone shall be met and no other damage or deterioration shall result after subjection to vibration tests specified herein.

3.16.2.2 Resistance to Impact.- The response requirements of the microphone shall be met and no other damage or deterioration shall result after subjection to impact tests specified herein.

3.16.2.3 Storage.- The microphone may be stored in any position for a period of one year.

3.16.3 Service Conditions (Special).- The microphone shall be protected from the harmful effects of the following:

3.16.3.1 Salt Sea Atmosphere.- The microphone shall be capable of withstanding the salt spray test specified in 4.2.2.11.

3.16.3.2 Insects.- Openings shall be screened wherever practicable, in order to protect parts whose function would be imparied by the presence of insects.

3.16.3.3 Fungus.- The equipment shall not be adversely affected by corrosion or fungus growth in tropical climates.

3.17 Finishes.- The parts of the microphone shall be finished so as to be resistent to corrosion, and there shall be no destructive corrosion after a 50-hour salt spray test in accordance with Specification QQ-M-151. Destructive corrosion shall be construed as being any type of corrosion which in any way interferes with mechanical and electrical performance. Bright finishes shall not be used on external parts. Where cleaning operations on metal parts are not specified in detail, they shall be in accordance with commercial practices which will not cause subsequent destructive corrosion. Aluminum alloys shall be anodized in accordance with Specification AN-QQ-A-696. Corrosion-resisting ferrous alloys shall be given a passivation treatment but need not receive any

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other protective coating or finish, unless such plating or finish is necessary or desirable for electrical or mechanical reasons. The plating on ordinary iron and steel shall be in accordance with the following:

a. Cadmium: QQ-P-416
b. Chromium: MIL-P-6871
c. Nickel: MIL-P-6859
d. Zinc: AN-P-32

3.18 Moisture Pockets.- Pockets, wells, traps and the like in which water and condensed moisture can collect when the microphone is in normal operating position shall be eliminated or properly drained.

3.19 Nameplate.- Each microphone shall have a nameplate mounted on or moulded in the rear of the housing on which shall be incorporated the following information:

M-15( )/UR
(Contract No.)
(Name of Manufacturer)*

The words in parenthesis will not appear in the marking on the microphone, but will be replaced by the data indicated.

3.20 Workmanship.- The microphone, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of the microphone shall be classified as follows:

4.1.1 Preproduction Tests.- Preproduction tests are those tests made on samples representative of the production microphones, to determine that the design of the production microphones meet the requirements of this specification.

4.1.2 Inspection Tests.- Inspection tests are those tests accomplished on microphones submitted for acceptance under contract.

4.2 Preproduction Tests:

4.2.1 Preproduction Test Sample.- Ten microphones, as specified in the contract, will be tested for design approval by the Procuring Agency, or when so specified in the contract, at the contractor's plant under the supervision of the Procuring Agency.

*The inclusion of the name of the manufacturer is optional.

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4.2.1.1 The preproduction samples shall be in all respects truly representative of the microphones the contractor proposes and expects to furnish in quantity production.

4.2.1.2 Parts Data.- The preproduction samples shall be accompanied by adequate identifying information for all parts and materials employed, including the following where applicable:

(1) The Government contract or order number involved.
(2) Prime Contractor's name and part number.
(3) Manufacturer's name and part number.
(4) In case the item or part has been approved previously, the date of approval, the name of the approving agency, and the previous contract or order number involved.

4.2.2 The preproduction test shall consist of the following tests and those specified under inspection test methods, subject to the test conditions specified herein:

4.2.2.1 Sound Pressure Meter.- The sound pressure meter specified in Drawing No. SC-D-10599, Figure 2, shall be used in setting up sound levels for performing the response measurements specified herein. The microphone of this sound pressure meter shall be a Western Electric 640-A or 640-AA condenser microphone, or a demonstrated equivalent. The sound pressure meter microphone shall be calibrated by the reciprocity method. The calibration of the microphone shall be done by one of the following Laboratories.

(1) National Bureau of Standards
(2) Bell Telephone Laboratories
(3) Electro-Acoustic Laboratory, Harvard University

4.2.2.2 Sound Source Calibration.- The equipment and test set-up shall be in accordance with Drawing No. SC-D-10599, Figure 1, or its approved equivalent, and shall be used to provide the sound input to the microphone under test for all response measurements. The test fixture employed to hold the microphone under test (to be used in lieu of Item 6 of Figure 1, Drawing No. SC-D-10599) shall be constructed so as to permit rotation of the microphone when mounted in the fixture (for conditioning purposes), and shall support the microphone in such fashion that the grid area of the microphone (with no surrounding baffle) is exposed to the sound source. The grid of the microphone under test shall be spaced 1/4 inch from the mouth of the sound source. The geometric center line of the grid shall coincide with the center line of the mouth of the sound source. The input to the driver unit required to produce a sound pressure of 115 decibels above a reference level of 0.00020 dyne per square centimeter at the grid of the microphone under test shall be determined for each 100 cycles per second increment from 300 to 1000 cycles per second, inclusive, each 200 cycles per second increment from 1200 to 4000 cycles per second, inclusive, and each 500 cycles per second increment from 4000 to 5000 cycles per second, inclusive. The grid of the 640-A condenser microphone shall be spaced 1/4 inch from the mouth of the sound source during calibration, and the center lines of the sound source mouth and the 640-A condenser microphone shall coincide. In calibrating the sound source, the condenser microphone shall

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be contained in a dummy microphone housing identical to the housing of the noise-canceling microphone to be tested. At each test frequency, the distortion in the output of the sound source shall be such that the level of the second harmonic is at least 35 decibels below the fundamental and the levels of the third and higher harmonics are at least 40 decibels below the fundamental.

4.2.2.3 Conditioning of Microphone.- Immediately prior to response tests, each microphone shall be conditioned using the test circuit in accordance with Drawing No. SC-D-5975, Figure 1, according to the following procedure.

4.2.2.3.1 The microphone shall be secured to the test fixture, and electrically connected to Jack J-1 of the test circuit.

4.2.2.3.2 Switch S-1 of the test circuit shall be closed (no microphone current) and Resistor R-1 shall be adjusted to produce a reading of 60 milliamperes on Meter M-1.

4.2.2.3.3 The microphone shall be rotated about its axis three times, through an arc of approximately 270 degrees and back, in two seconds. The microphone shall be in the position at which it is to be tested at the completion of this rotation procedure.

4.2.2.3.4 Switch S-1 shall be opened, approximately three seconds shall be allowed for current stabilization, then the conditioning sweep tone described in 4.2.2.3.5 and 4.2.2.3.6 shall be applied. The voltage of the conditioning sweep tone shall have been previously determined as follows:

4.2.2.3.4.1 The oscillator shall be set at that frequency which required the lowest driver input voltage to produce a sound pressure of 115 decibels.

4.2.2.3.4.2 The driver input voltage shall be adjusted at that frequency to four times the voltage required to give a 115 decibel sound level (the maximum sound pressure developed during conditioning shall, therefor, not exceed 127 decibels.)

4.2.2.3.4.3 The oscillator frequency shall be changed to 1000 cycles per second without disturbing the output control of the oscillator. The voltage reading on Output Meter 2 of Drawing No. SC-D-10599, Figure 1, shall then be taken as the 1000 cycles per second reference conditioning voltage.

4.2.2.3.5 The driver input voltage shall be adjusted at 1000 cycles per second to the value required for conditioning (see 4.2.2.3.4.3).

4.2.2.3.6 The oscillator frequency shall be continuously varied through the 1000 to 3000 to 1000 cycles per second audio frequency range three times at a rate of approximately one complete sweep in two seconds.

4.2.2.3.7 At the conclusion of the third sweep cycle, the oscillator output voltage shall be gradually reduced to zero.

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4.2.2.3.8 Approximately two seconds shall elapse before starting response measurements on a specific test run.

CAUTION: The microphone shall not be disturbed by loud clicks from the driver unit, noises, vibrations, or accidental bumps during "conditioning", or measurements of response, resistance or "burning".

4.2.2.4 Response.- After the microphone under test has been conditioned in accordance with the above procedure, the sound level, determined as specified under 4.2.2.2 shall be applied. The response (output level) of the microphone shall be determined for each test-frequency. Measurements shall start at 300 cycles per second and progress through each successively higher test frequency up to 5000 cycles per second. The sound output from the driver shall be left on when changing the oscillator from one test frequency to the next, but the sound input to the microphone shall not exceed 115 decibels in effecting the changes in frequency. This shall not be construed as prohibiting the use of push-button frequency control in production testing providing that substantially equivalent results are obtained.

4.2.2.5 Microphone Resistance.- The microphone resistance shall be measured at the time the 1000 cycles per second response is being determined, by observing the reading of Meter M-3 of Figure 1, Drawing No. SC-D-5975, with Switch S-4 closed and Switch S-1 open. The microphone resistance shall be calculated from the equation $R = \frac{400E}{24-E}$ where R is the resistance of the microphone in ohms and E is the reading of Meter M-3 in volts.

4.2.2.6 Word Articulation Tests:

4.2.2.6.1 Ambient Noise.- The articulation tests shall be performed under noise-to-quiet (N-Q) conditions. Noise-to-quiet signifies that the caller or announcer is in a noise field, and the listeners are in quiet. The noise spectrum shall approximate that shown on Drawing No. 49B12538. The overall level of the simulated airplane noise shall be 120 decibels above 0.00020 dyne per square centimeter.

4.2.2.6.2 Test Team.- At least two experienced callers, and at least four experienced listeners shall make up the articulation test team. The callers shall have demonstrated the ability to use standard American speech proficiently, to enunciate clearly, and to talk well under extremely noisy conditions. The listeners shall have essentially normal hearing. Callers shall not be used as listeners in the articulation tests.

4.2.2.6.3 Test Lists.- The word lists used in the articulation tests shall be made up of single syllable, phonetically-balanced, commonly used words, and shall not be read in alphabetical sequence. The word lists shall be of approximately equal average difficulty and have an equal range of difficulty. Acceptable word lists of this type are the "PB" lists in

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Appendix II of Office of Scientific Research and Development Report No. 3802, titled: "Articulation Testing Methods II." However, those lists require rearrangement to eliminate alphabetical sequence.

4.2.2.6.4 Number of Tests.- At least 500 words shall be called over each test microphone by each caller.

4.2.2.6.5 Test Equipment.- The microphone shall be tested using the equipment shown in Drawing No. SC-D-5975, Figure 6.

4.2.2.7 Response Curves.- The response curves of the microphone based on the contractor's test data shall be plotted on twenty divisions per inch (total 120 divisions) by two 4-1/2 inch cycle, semi-logarithmic graph paper. The axis of ordinates shall cover the range from plus 10 to plus 70 decibels; each vertical division shall be 0.5 decibel. Resistance and burning values for the microphones also shall be shown on the curve sheets.

4.2.2.8 Temperature Tests.- To determine compliance with the temperature requirements of this specification, the microphone shall be subjected to temperature cycling tests and temperature extremes as outlined below.

4.2.2.8.1 Temperature Cycling, Equipment Non-Operating:

4.2.2.8.1.1 The microphone shall be placed in an ambient temperature of 85°C (185°F) for 60 minutes.

4.2.2.8.1.2 The microphone shall be removed from the hot chamber and allowed to remain in normal room temperature for 15 minutes.

4.2.2.8.1.3 The microphone shall be placed in ambient temperature of -65°C (-85°F) for 60 minutes.

4.2.2.8.1.4 The microphone shall be removed from the cold chamber and allowed to remain at normal room temperature for 15 minutes.

4.2.2.8.1.5 The above procedure shall be repeated until the microphone has been exposed to each extreme temperature for a total of ten hours; that is, for ten complete hot and cold cycles.

4.2.2.8.1.6 The microphone shall be removed from the cold chamber after the tenth cold exposure, and allowed at least one hour to regain room temperature before making response tests.

4.2.2.8.2 Temperature Extremes, Operating:

4.2.2.8.2.1 Cold Exposure:

4.2.2.8.2.1.1 Expose the microphone to an ambient temperature of -55°C (-67°F) for 2 hours.

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4.2.2.8.2.1.2 Remove the microphone from the cold chamber and immediately measure its response under normal room temperature conditions.

4.2.2.8.2.2 Heat Exposure:

4.2.2.8.2.2.1 The microphone shall be exposed to an ambient temperature of 71°C (160°F) for 2 hours, then removed from the hot chamber and the response measured immediately under normal room temperature conditions.

4.2.2.9 Altitude Tests.- To determine compliance with the pressure requirement of this specification, the microphone shall be subjected to the tests described below.

4.2.2.9.1 Pressure Cycling.- The microphone shall be subjected to three pressure cycles. Each pressure cycle shall consist of one "ascent" to 50,000 feet, approximately 3.4 inches of mercury, remaining at that altitude for at least 1 minute, and one "descent" to ground level. The rate of ascent and descent shall be approximately 5000 feet per minute. The response characteristic of the microphone shall then be measured to determine if this treatment resulted in permanent or serious changes in response curve shape and output level.

4.2.2.9.2 Response at Simulated Altitude of 15,000 Feet.- The microphone shall be placed in an altitude chamber and the pressure inside the chamber reduced to that corresponding to an altitude of 15,000 feet, approximately 16.7 inches of mercury. The response of the microphone shall be measured at that altitude. Also, the response at 1000 cycles per second shall be measured while the pressure is being varied at the rate of 5000 feet per minute (up or down).

4.2.2.9.3 Talk-Out Test at Altitude.- The microphone shall be subjected to a talk-out test at an altitude of 10,000 feet (20.3 inches of mercury). The test circuit shall be in accordance with Drawing No. SC-D-5975, Figure 1. The test shall be conducted with a noise field in the altitude chamber in accordance with Drawing No. 49B12538, at an overall sound pressure level of 120 decibels.

4.2.2.10 Humidity.- The microphone shall be exposed to the humidity cycling specified in Specification JAN-M-745 for 120 hours (five 24-hour cycles). The response characteristic shall be measured immediately after removal from the humidity chamber, and again after allowing the microphone to remain under normal room conditions for 24 hours.

4.2.2.11 Salt Spray.- The microphone shall be exposed for 100 hours to the salt spray tests specified in Specification QQ-M-151. The response characteristic shall be measured immediately after removal from the salt spray chamber, and again after allowing the microphone unit to remain under normal room conditions for 24 hours.

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4.2.2.12 Fungus.- The microphone shall be tested for fungus resistance in accordance with Specification MIL-E-5272.

4.2.2.13 Vibration.- The microphone shall be mounted on a vibration table in a manner similar to a service installation, including use of any mounting or other holding device. The microphone shall be vibrated with the frequency varying between 10 and 55 cycles per second at an amplitude of 0.06 inch total excursion, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately one minute. Application of the vibration shall be in each of the following directions:

(1) Horizontally, parallel to the major horizontal axis of the microphone for a period of 90 minutes.
(2) Horizontally, at right angles to the major horizontal axis, for a period of 90 minutes.
(3) Vertically, for a period of 90 minutes.

The response of the microphone shall be measured after the vibration test.

4.2.2.14 Cordage Flexing Test.- Five sample lengths of cordage, each approximately 1-1/2 feet long, shall be clamped and suspended through holes in a rectangular (1/2 x 5/8 inch) metal bar in the manner described below:

(1) The clamping point shall be immediately above the metal bar.
(2) The cordage shall be clamped in such a manner that it will not turn in the hole during the flexing test.
(3) The length of the through holes in the metal bar shall be 5/8 inch. The diameter of the through holes shall be not greater than 0.295 inch.
(4) The through holes in the metal bar shall incorporate a 1/8 inch radius at the end from which the cord emerges (bending point).
(5) A knot shall be tied near the free end of the cord and a one-pound weight shall be hung from the knot.

The three conductors of each sample length of cordage, a small low current drain 6-volt lamp, and a suitable power source shall be connected in series to indicate electrical continuity. The metal bar, supported horizontally, shall be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute.

4.3 Inspection Tests:

4.3.1 Inspection Samples:

4.3.1.1 One microphone shall be selected at random from each lot of 100 or fraction thereof on the order and subjected to the sampling tests. These tests shall be in addition to the individual tests specified herein.

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4.3.1.2 When tests are specified on a quantity of microphones that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional microphones of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the microphone.

4.3.2 Unless otherwise specified, contractor's records of all inspection work, giving the results of tests required to determine compliance with the requirements specified herein, shall be kept complete and shall be available to the Government representative at all times. The tests shall be accomplished on articles to be supplied on the contract or order. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.3.3 Inspection of the microphone shall consist of the tests specified under inspection test methods, subject to the test conditions specified herein.

4.4 Test Conditions.- Unless otherwise specified or indicated, the microphone shall be tested under the following conditions:

4.4.1 Temperature.- 20 to 35°C (68 to 95°F)

4.4.2 Pressure.- Normal atmospheric

4.4.3 Humidity.- 25 to 90 percent relative

4.4.4 Position.- The face of the microphone shall be in a vertical plane.

4.5 Inspection Test Methods:

4.5.1 Individual Tests.- Each microphone shall be subjected to the following tests:

4.5.1.1 Response Measurements.- Prior to making response measurements, the microphone shall be conditioned as specified herein or according to some other specifically approved method which will produce equivalent results. Each microphone shall be subjected to response tests at not less than six approved frequencies, such as 400, 1000, 2000, 3000, 4000 and 5000 cycles per second. The specified frequencies to be used will be determined by the Procuring Agency on the basis of tests made on the preproduction samples and information gained in the initial stages of production. Sweep or band frequency tests may be employed provided the contractor demonstrates the suitability of such tests to the Procuring Agency. The method of testing and the selected frequencies shall be such that they will truly indicate the overall response characteristics and output level of the microphone.

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4.5.1.2 Microphone Resistance.- The resistance of each microphone shall be measured.

4.5.1.3 Noise-Cancellation.- The contractor shall provide adequate means for testing the noise-cancellation of production microphones and shall insure that the noise cancellation properties of production microphones are equivalent to those of an approved preproduction sample.

4.5.1.4 Talk-Out Test.- Each microphone complete with microphone cover shall be given a talk-out test in a noise booth. The test circuit shall be in accordance with Drawing No. SC-D-5975, Figure 1. The spectrum of the noise field in the booth shall be approximately that shown on Drawing No. 49B12538 and shall have an overall sound pressure level of 120 decibels with the test operator in the booth. The microphone shall be talked into while in various positions, that is, horizontal, vertical, and so forth. During this test an observer shall listen to the caller's voice using a suitable headset plugged into the alternating-current output jack of the electronic voltmeter of the test circuit. Each microphone shall perform satisfactorily in this test and shall show no pronounced differences in loudness of speech quality when operated in the various positions.

4.5.2 Sampling Tests.- The samples shall be subject to such inspection and tests as are necessary to determine complete compliance with the requirements of this specification. The sampling tests, in general, shall consist of the following:

4.5.2.1 Complete Response Tests.- The samples shall be subjected to complete response tests in accordance with 4.2.2.4, tests being made at each 100 cycles per second frequency increment from 300 to 1000 cycles per second, inclusive, and at each 200 cycles per second increment from 1200 to 4000 cycles per second, inclusive, and at 4500 and 5000 cycles per second.

4.5.2.2 Burning.- The output of the microphone due to burning shall be measured in the circuit shown in Drawing No. SC-D-5975, Figure 1, as follows:

4.5.2.2.1 After the microphone has been conditioned according to the procedure given in 4.2.2.3, it shall remain still and without signal input for 30 seconds. The maximum output, as indicated on Meter M-2 of the test circuit during the 10-second interval immediately following the 30-second period, shall be the measure of the burning.

4.5.2.3 Artificial Aging.- The microphone, connected in the test circuit in accordance with Drawing No. SC-D-5975, Figure 3, shall be dropped onto a flat wooden block in such a manner that the face of the microphone hits as squarely as possible on the block. The weight of the device to which the microphone is secured shall be such as to produce a total combined weight of 1.0 pound. The microphone shall be dropped through a distance of 1-1/2 inches with an acceleration approximately equal to that due to gravity, at a rate of 35 times per minute. The period during which the electrical circuit is operative

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shall start at the instant the microphone begins to drop and shall continue for 25 percent of the time interval required for the microphone to regain the position from which it starts its drop. A microphone which has been dropped 10,000 times shall be considered an aged microphone.

4.5.2.4 Anti-Packing.- The microphone shall be tested for surges in the circuit of Drawing No. SC-D-5975, Figure 5. Tests for overload current and extended operation shall be made in accordance with the circuit shown in Drawing No. SC-D-5975, Figure 1.

4.5.2.5 Resistance to Impact.- The microphone shall withstand 20 drops from a height of five feet to a concrete floor without structural failure. Furthermore, the transmission characteristics of the microphone shall not be seriously affected as a result of this test.

5. PREPARATION FOR DELIVERY

5.1 General.- The packaging, packing and marking shall be in accordance with Method II of Specification No. 7200.

6. NOTES

6.1 Use.- Carbon Microphone M-15( )/UR is intended to be used in Military aircraft.

6.2 Ordering Data.- Invitations for bids, contracts and purchase orders should state the conditions for the following.

6.2.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that ten microphones will be required as First Articles (or preproduction samples) and that these First Articles (or preproduction samples) will be subjected to the Preproduction Tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

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